

ICBHI 2017 Respiratory Sound Database

920 recordings · 126 patients · 5.5 h · 4 stethoscopes
one .txt per recording: cycle start, end, crackle, wheeze

ICBHI_challenge_train_test.txt

the released official split file
assigns RECORDINGS, not patients — two patients straddle it

AUDITED SPLIT LOADER (contribution 1)

raise if the split file is absent — never a silent identifier fallback
detect patients on both sides (156, 218) → reassign every recording to train
device-suffix-safe join: 226_1b1_PL_sc_Meditron is released as ..._LittC2SE
assert the train and test patient sets are disjoint before returning

TRAIN partition

79 patients · 4,251 cycles

TEST partition

47 patients · 2,636 cycles

training path

evaluation path

SHARED PREPROCESSING (identical for both partitions)

load 16 kHz mono, crop to the annotated cycle start/end
cyclic tiling to 8.0 s (Eq. 1) — 99.8 % of cycles are shorter
log-mel 128×801: $n_{\text{fft}} 1024$, win 400, hop 160, 50–2000 Hz (Eq. 2)
per-spectrogram min-max to [0, 1] → 1 → 3 channels + ImageNet mean/std

test cycles: no augmentation

SpecAugment (training split only)

2 freq masks ≤ 24 bins, 2 time masks ≤ 80 frames
resampled every epoch (Eq. 5)

not in the baseline

band-pass P1 · denoise P2
amplitude normalisation P3
each is an ADDITIVE ablation row

BACKBONE (one of four, ImageNet-1k) + shared classifier head

MobileNetV2 2.23 M | Swin-T 27.5 M | DeiT-S 21.7 M | ViT-B/16 85.8 M
global average pool → dropout 0.3 → Linear(→ 4 logits)
inverse-frequency class-weighted cross-entropy, unit mean (Eq. 4)

checkpoint selection
(Fault 5, Sec. III-C):
keep argmax official
score, not min loss

4-class prediction per test cycle → raw 4×4 confusion matrix

OFFICIAL SCORING MODULE (contribution 2)

commit the raw 4×4 confusion matrix verbatim — every score stays recomputable
ICBHI = $(Se + Sp)/2$ with the three abnormal classes POOLED (Eq. 6)
the macro variant is still computed, but reported under its own name (Eq. 7)
patient-level percentile bootstrap, $B = 1000$ · paired McNemar and bootstrap

results_M*.json → every table in Section III (best model: ICBHI 0.5602)